The China Syndrome: Local Labor Market Effects of Import Competition in the US

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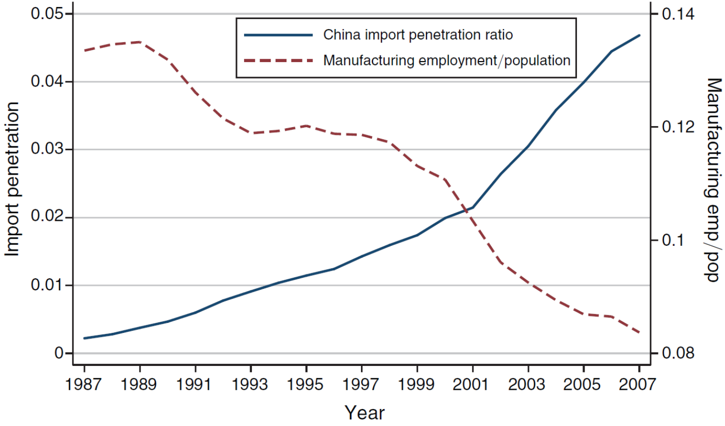
China:

- Market-oriented reforms since the 1980s
- Decentralisation, Special Economic Zones, consolidations of SOEs and etc.
- Accession to the WTO and China as 'The World's Factory'

The US:

- Technological change, rise of multinationals and outsourcing
- Significant increase of bilateral trade with China but one-sided
- Deindustrialisation (Rust Belt) & decline in manufacturing employment
- Resentments towards globalisation and free trade



Import penetration: imports divided by total US expenditure on goods

Key Issues:

- Did rising imports from China contributed to the decline in US manufacturing employment?
- What about other labour market effects? E.g. population growth & wages
- Is there an increase in governmental transfer payments (welfare programmes) as a result of Chinese import shock?
- Overall welfare analysis? Gains vs losses from trade with China
- Empirical challenges: correlation $\neq$ causation

Monopolistic Competition Model (Helpman-Krugman):

- The US is separated into 722 communiting zones (CZ) each representing a ‘small open economy’
- Every CZ produces two traded goods and a homogeneous non-traded goods
- Productivity is given
- Only labour is used in production and there is no labour migration across CZ (hence short to medium run effects)
- Cobb-Douglas utility function: $\gamma/2$ of expenditure goes to each of the two trade goods and $1 - \gamma$ goes to non-traded goods

Model Solution

Deriving the impacts of Chinese trade shocks:

$$\hat{W}_i = \sum_j c_{ij} \frac{L_{ij}}{L_{Ni}} \left[\theta_{ijC} \hat{E}_{Cj} - \sum_k \theta_{ijk} \phi_{Cjk} \hat{A}_{Cj} \right]$$

$$\hat{L}_{Ti} = \rho_i \sum_j c_{ij} \frac{L_{ij}}{L_{Ti}} \left[\theta_{ijC} \hat{E}_{Cj} - \sum_k \theta_{ijk} \phi_{Cjk} \hat{A}_{Cj} \right]$$

$$\hat{L}_{Ni} = \rho_i \sum_j c_{ij} \frac{L_{ij}}{L_{Ni}} \left[-\theta_{ijC} \hat{E}_{Cj} + \sum_k \theta_{ijk} \phi_{Cjk} \hat{A}_{Cj} \right]$$

Trade with China impacts the US labour market through two channels:

- Increased competition from China's export-supply capacity ($\hat{A}_{Cj}$)
- Increased demand for US goods from China ($\hat{E}_{Cj}$)

Model Solution

$$\hat{L}_{Ti} = \rho_i \sum_j c_{ij} \frac{L_{ij}}{L_{Ti}} \left[\theta_{ijC} \hat{E}_{Cj} - \sum_k \theta_{ijk} \phi_{Cjk} \hat{A}_{Cj} \right]$$

Other parameters:

- Share of the current-account deficit in total expenditure ($p_i < 0$)
- General-equilibrium scaling factor ($c_{ij} > 0$)
- Share of sector j labour ($\frac{L_{ij}}{L_{Ti}}$)
- Share of US output exported to China ($\theta_{ijC} \equiv \frac{X_{ijC}}{X_{ij}}$)
- Share of US output exported to market k ($\theta_{ijk} \equiv \frac{X_{ijk}}{X_{ij}}$)
- Share of Chinese imports in market k's total expenditure ($\phi_{Cjk} \equiv \frac{M_{kjC}}{E_{kj}}$)

Model Solution

Simplifying for empirical analysis:

$$\hat{L}_{Ti} = \rho_i \sum_j c_{ij} \frac{L_{ij}}{L_{Ti}} \left[\theta_{ijC} \hat{E}_{Cj} - \sum_k \theta_{ijk} \phi_{Cjk} \hat{A}_{Cj} \right]$$

$$= \alpha \sum_j \frac{L_{ij}}{L_{Ti}} \left[\theta_{ijC} \hat{E}_{Cj} - \sum_k \theta_{ijk} \phi_{Cjk} \hat{A}_{Cj} \right] \qquad (\rho_i c_{ij} = \alpha)$$

$$= -\alpha \sum_j \frac{L_{ij}}{L_{Ti}} \theta_{ijk} \phi_{Cjk} \hat{A}_{Cj} \qquad (\hat{E}_{Cj} = 0)$$

$$= -\alpha \sum_j \frac{L_{ij}}{L_{Ti}} \frac{X_{ijU}}{X_{ij}} \frac{M_{CjU}}{E_{Uj}} \hat{A}_{Cj}$$

$$\approx -\tilde{\alpha} \sum_j \frac{L_{ij}}{L_{Uj}} \frac{M_{CjU} \hat{A}_{Cj}}{L_{Ti}} \qquad \left(\alpha \frac{L_{ij}}{X_{ij}} = \tilde{\alpha}; \frac{X_{ijU}}{E_{Uj}} \approx \frac{L_{ij}}{L_{Uj}} \right)$$

Empirical Proxy

$$\hat{L}_{Ti} \approx -\tilde{\alpha} \sum_j \frac{L_{ij}}{L_{Uj}} \frac{M_{CjU} \hat{A}_{Cj}}{L_{Ti}}$$

Define change in Chinese import exposure per worker in CZ i in time t as:

$$\Delta IPW_{uit} = \sum_j \frac{L_{ijt}}{L_{ujt}} \frac{\Delta M_{ucjt}}{L_{it}}$$

- Observed change in US imports from China in industry j (ΔM_{ucjt})
- Two other sources of variations:

1. Specialization in import-intensive industries within local manufacturing ($\frac{L_{ijt}}{L_{ujt}}$)
2. Differential concentration of employment in manufacturing versus nonmanufacturing activities ($\frac{L_{ijt}}{L_{it}}$)

$$\text{Model:} \quad \hat{L}_{Ti} \approx -\tilde{\alpha} \sum_j \frac{L_{ij}}{L_{Uj}} \frac{M_{CjU}\hat{A}_{Cj}}{L_{Ti}}$$

$$\text{Proxy:} \quad \Delta IPW_{uit} = \sum_j \frac{L_{ijt}}{L_{ujt}} \frac{\Delta M_{ucjt}}{L_{it}}$$

But does $M_{CjU}\hat{A}_{Cj} = \Delta M_{ucjt}$?

- Realized US imports from China (ΔM_{ucjt}) may be correlated with unobserved industry import demand shocks
- Employment is positively correlated with demand shocks + negative coefficients $\Rightarrow$ negative bias on OLS estimates
- CZ measurement errors $\Rightarrow$ downward bias towards zero

Identification Strategy

IV Approach:

$$\Delta IPW_{oit} = \sum_j \frac{L_{ijt-1}}{L_{ujt-1}} \frac{\Delta M_{ocjt}}{L_{it-1}}$$

- Same Chinese exposure variable is constructed for eight other developed countries
- Ten-year-lagged employment data $(t-1)$ to mitigate potential simultaneity bias from anticipated China trade

Identification Assumption: 'common within-industry component of rising Chinese imports to the US and other high-income countries stems from China's rising comparative advantage and (or) falling trade sector'

First Differenced 2SLS:

$$\Delta \widehat{IPW}_{uit} = \hat{\delta}_1 \Delta IPW_{oit} + \epsilon_{it} \qquad \text{(First Stage)}$$

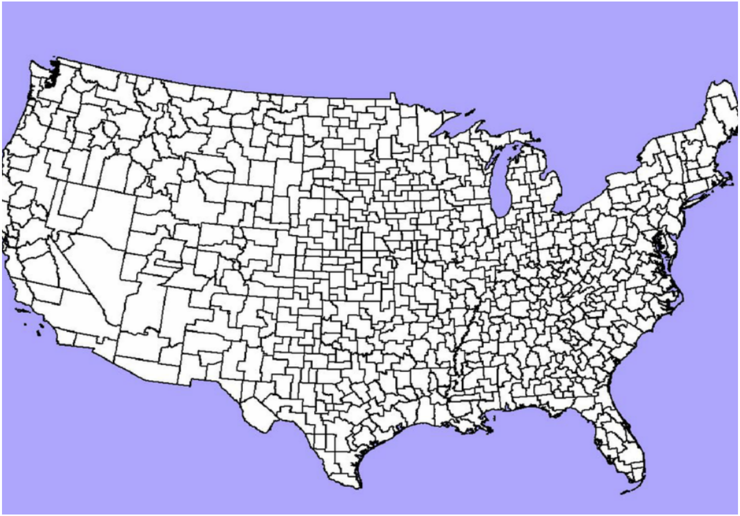
$$\Delta L^m_{it} = \gamma_t + \beta_1 \Delta \widehat{IPW}_{uit} + X'_{it}\beta_2 + e_{it} \qquad \text{(Second Stage)}$$

- ΔL^m_{it}: changes in the manufacturing employment share in CZ i in decade t
- γ_t: time dummy for each decade
- X_{it}: control vector (e.g. start-of-decade labor force and demographic composition)
- Standard errors are clustered at the state level
- First difference estimator $\Rightarrow$ CZ fixed-effect

Data Sources

- Classification of Commuting Zones (CZs) is based on the **1990 census**
- Trade data comes from the **UN Comtrade Database**
- Manufacturing industry employment structure by CZs is derived from the **County Business Patterns data**
- Data on population, employment, and wage structure by education, age, and gender in CZs comes from the **Census Integrated Public Use Micro Samples** and the **American Community Survey** (ACS)
- Federal and state transfer payments to CZ residents comes from the **Bureau of Economic Analysis and the Social Security Administration**

Commuting Zones



Benchmark Specification

TABLE 2—IMPORTS FROM CHINA AND CHANGE OF MANUFACTURING EMPLOYMENT IN CZS, 1970–2007: 2SLS ESTIMATES

Dependent variable: 10 × annual change in manufacturing emp/working-age pop (in % pts)

	I. 1990–2007			II. 1970–1990 (pre-exposure)		
	1990–2000 (1)	2000–2007 (2)	1990–2007 (3)	1970–1980 (4)	1980–1990 (5)	1970–1990 (6)
(Δ current period imports from China to US)/worker	−0.89*** (0.18)	−0.72*** (0.06)	−0.75*** (0.07)			
(Δ future period imports from China to US)/worker				0.43*** (0.15)	−0.13 (0.13)	0.15 (0.09)

Notes: $N = 722$, except $N = 1{,}444$ in stacked first difference models of columns 3 and 6. The variable "future period imports" is defined as the average of the growth of a CZ's import exposure during the periods 1990–2000 and 2000–2007. All regressions include a constant and the models in columns 3 and 6 include a time dummy. Robust standard errors in parentheses are clustered on state. Models are weighted by start of period CZ share of national population.

***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

- Interpretation: \$1,000 exogenous decadal rise in a CZ's import exposure per worker is predicted to reduce its manufacturing employment share by 0.75 percentage point in the period of 1900-2007

Falsification Test

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- Falsification test: regressing past changes in the manufacturing employment share on future changes in import exposure
- No evidence of reverse causality

TABLE 3—IMPORTS FROM CHINA AND CHANGE OF MANUFACTURING EMPLOYMENT IN CZs, 1990–2007: 2SLS ESTIMATES

Dependent variable: 10 × annual change in manufacturing emp/working-age pop (in % pts)

	I. 1990–2007 stacked first differences					
	(1)	(2)	(3)	(4)	(5)	(6)
(Δ imports from China to US)/worker	$\boxed{-0.746^{***}}$	-0.610^{***}	-0.538^{***}	-0.508^{***}	-0.562^{***}	-0.596^{***}
	$\boxed{(0.068)}$	(0.094)	(0.091)	(0.081)	(0.096)	(0.099)
Percentage of employment in manufacturing$_{-1}$		−0.035	-0.052^{***}	-0.061^{***}	-0.056^{***}	-0.040^{***}
		(0.022)	(0.020)	(0.017)	(0.016)	(0.013)
Percentage of college-educated population$_{-1}$				−0.008		0.013
				(0.016)		(0.012)
Percentage of foreign-born population$_{-1}$				−0.007		0.030^{***}
				(0.008)		(0.011)
Percentage of employment among women$_{-1}$				-0.054^{**}		−0.006
				(0.025)		(0.024)
Percentage of employment in routine occupations$_{-1}$					-0.230^{***}	-0.245^{***}
					(0.063)	(0.064)
Average offshorability index of occupations$_{-1}$					0.244	−0.059
					(0.252)	(0.237)
Census division dummies	No	No	Yes	Yes	Yes	Yes
	II. 2SLS first stage estimates					
(Δ imports from China to OTH)/worker	0.792^{***}	0.664^{***}	0.652^{***}	0.635^{***}	0.638^{***}	0.631^{***}
	(0.079)	(0.086)	(0.090)	(0.090)	(0.087)	(0.087)
R^2	0.54	0.57	0.58	0.58	0.58	0.58

- The benchmark results (green box) is robust across all specifications

Augmented Specification (Robustness Checks)

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- Controlling for overall trend decline in US manufacturing

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- Controlling for demographic structure

Augmented Specification (Robustness Checks)

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	(0.079)	(0.086)	(0.090)	(0.090)	(0.087)	(0.087)
R^2	0.54	0.57	0.58	0.58	0.58	0.58

- Controlling for relative risk of automation and offshoring

Changes in Working-age Population

TABLE 4—IMPORTS FROM CHINA AND CHANGE OF WORKING-AGE POPULATION IN CZ, 1990–2007: 2SLS ESTIMATES

Dependent variables: Ten-year equivalent changes in log population counts (in log pts)

	I. By education level			II. By age group		
	All (1)	College (2)	Noncollege (3)	Age 16–34 (4)	Age 35–49 (5)	Age 50–64 (6)
Panel A. No census division dummies or other controls						
(Δ imports from China to US)/worker	−1.031**	−0.360	−1.097**	−1.299	−0.615	−1.127***
	(0.503)	(0.660)	(0.488)	(0.826)	(0.572)	(0.422)
R^2	—	0.03	0.00	0.17	0.59	0.22
Panel B. Controlling for census division dummies						
(Δ imports from China to US)/worker	−0.355	0.147	−0.240	−0.408	−0.045	−0.549
	(0.513)	(0.619)	(0.519)	(0.953)	(0.474)	(0.450)
R^2	0.36	0.29	0.45	0.42	0.68	0.46
Panel C. Full controls						
(Δ imports from China to US)/worker	−0.050	−0.026	−0.047	−0.138	0.367	−0.138
	(0.746)	(0.685)	(0.823)	(1.190)	(0.560)	(0.651)
R^2	0.42	0.35	0.52	0.44	0.75	0.60

- Regressions are analogous to manufacturing employment share (only changing the dependent variable)
- Did Chinese trade shocks cause a decline in CZ's working population? – No evidence when accounting for controls (Panel C)

Changes in Working-age Population

Potential Reasons:

- Chinese trade shocks are too small to change the demographic structure of CZs
- Goods markets are sufficiently well integrated nationally to adjust for Chinese trade shocks
- Sluggish US labour mobility (e.g. housing prices, lacking transferable skills)

No change in working-age population + decline in manufacturing employment

⇒ Higher unemployment? Higher employment in other sectors? Labour force exit?

TABLE 5—IMPORTS FROM CHINA AND EMPLOYMENT STATUS OF WORKING-AGE POPULATION WITHIN CZs, 1990–2007: 2SLS ESTIMATES
Dependent variables: Ten-year equivalent changes in log population counts and population shares by employment status

	Mfg emp (1)	Non-mfg emp (2)	Unemp (3)	NILF (4)	SSDI receipt (5)
Panel A. 100 × log change in population counts					
(Δ imports from China to US)/worker	−4.231***	−0.274	4.921***	2.058*	1.466***
	(1.047)	(0.651)	(1.128)	(1.080)	(0.557)
Panel B. Change in population shares					
All education levels					
(Δ imports from China to US)/worker	−0.596***	−0.178	0.221***	0.553***	0.076***
	(0.099)	(0.137)	(0.058)	(0.150)	(0.028)
College education					
(Δ imports from China to US)/worker	−0.592***	0.168	0.119***	0.304***	—
	(0.125)	(0.122)	(0.039)	(0.113)	
No college education					
(Δ imports from China to US)/worker	−0.581***	−0.531***	0.282***	0.831***	—
	(0.095)	(0.203)	(0.085)	(0.211)	

- Working-age population is split into 4 mutually exclusive categories
- Overall, decline in manufacturing employment is met with rising unemployment and labour force exit

Within CZ Labour Adjustment

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	(0.095)	(0.203)	(0.085)	(0.211)	

- Non-college educated labour are much more adversely impacted by the Chinese trade shock, even in non-manufacturing sectors

Effect on wages

TABLE 6—IMPORTS FROM CHINA AND WAGE CHANGES WITHIN CZs, 1990–2007: 2SLS ESTIMATES

Dependent variable: Ten-year equivalent change in average log weekly wage (in log pts)

	All workers (1)	Males (2)	Females (3)
Panel A. All education levels			
(Δ imports from China to US)/worker	−0.759***	−0.892***	−0.614***
	(0.253)	(0.294)	(0.237)
R^2	0.56	0.44	0.69
Panel B. College education			
(Δ imports from China to US)/worker	−0.757**	−0.991***	−0.525*
	(0.308)	(0.374)	(0.279)
R^2	0.52	0.39	0.63
Panel C. No college education			
(Δ imports from China to US)/worker	−0.814***	−0.703***	−1.116***
	(0.236)	(0.250)	(0.278)
R^2	0.52	0.45	0.59

- Potential upward bias: workers with lower ability and earnings are more likely to lose employment
- Despite this, there are significant negative effect on wages across gender and educational level

Heterogeneous Wage & Employment Effects Across Sectors

TABLE 7—COMPARING EMPLOYMENT AND WAGE CHANGES IN MANUFACTURING AND OUTSIDE MANUFACTURING, 1990–2007: 2SLS ESTIMATES

Dependent variables: Ten-year equivalent changes in log workers and average log weekly wages

	I. Manufacturing sector			II. Nonmanufacturing		
	All workers (1)	College (2)	Noncollege (3)	All workers (4)	College (5)	Noncollege (6)
Panel A. Log change in number of workers						
(Δ imports from China to US)/worker	−4.231***	−3.992***	−4.493***	−0.274	0.291	−1.037
	(1.047)	(1.181)	(1.243)	(0.651)	(0.590)	(0.764)
R^2	0.31	0.30	0.34	0.35	0.29	0.53
Panel B. Change in average log wage						
(Δ imports from China to US)/worker	0.150	0.458	−0.101	−0.761***	−0.743**	−0.822***
	(0.482)	(0.340)	(0.369)	(0.260)	(0.297)	(0.246)
R^2	0.22	0.21	0.33	0.60	0.54	0.51

- Manufacturing sector: negative employment effect but no evidence for changes in wages
- Non-manufacturing sector: no evidence for employment effect but negative wage effects

Public Transfer Payments

TABLE 8—IMPORTS FROM CHINA AND CHANGE OF GOVERNMENT TRANSFER RECEIPTS IN CZS, 1990–2007: 2SLS ESTIMATES

Dep vars: Ten-year equivalent log and dollar change of annual transfer receipts per capita (in log pts and US$)

	Total individual transfers (1)	TAA benefits (2)	Unemployment benefits (3)	SSA retirement benefits (4)	SSA disability benefits (5)	Medical benefits (6)	Federal income assist (7)	Educ/training assist (8)
Panel A. Log change of transfer receipts per capita								
(Δ imports from China to US)/worker	1.01***	14.41*	3.46*	0.72*	1.96***	0.54	3.04***	2.78**
	(0.33)	(7.59)	(1.87)	(0.38)	(0.69)	(0.49)	(0.96)	(1.32)
R^2	0.57	0.28	0.48	0.36	0.32	0.27	0.54	0.33
Panel B. Dollar change of transfer receipts per capita								
(Δ imports from China to US)/worker	57.73***	0.23	3.42	10.00*	8.40***	18.27	7.20***	3.71***
	(18.41)	(0.17)	(2.26)	(5.45)	(2.21)	(11.84)	(2.35)	(1.44)
R^2	0.75	0.28	0.41	0.47	0.63	0.66	0.53	0.37

- Panel A investigates the uptakes of various welfare programmes
- Significant uptakes in Trade Adjustment Assistance (TAA), this is expected

TABLE 8—IMPORTS FROM CHINA AND CHANGE OF GOVERNMENT TRANSFER RECEIPTS IN CZS, 1990–2007: 2SLS ESTIMATES

Dep vars: Ten-year equivalent log and dollar change of annual transfer receipts per capita (in log pts and US$)

	Total individual transfers (1)	TAA benefits (2)	Unem-ployment benefits (3)	SSA retirement benefits (4)	SSA disability benefits (5)	Medical benefits (6)	Federal income assist (7)	Educ/training assist (8)
Panel A. Log change of transfer receipts per capita								
(Δ imports from China to US)/worker	1.01***	14.41*	3.46*	0.72*	1.96***	0.54	3.04***	2.78**
	(0.33)	(7.59)	(1.87)	(0.38)	(0.69)	(0.49)	(0.96)	(1.32)
R^2	0.57	0.28	0.48	0.36	0.32	0.27	0.54	0.33
Panel B. Dollar change of transfer receipts per capita								
(Δ imports from China to US)/worker	57.73***	0.23	3.42	10.00*	8.40***	18.27	7.20***	3.71***
	(18.41)	(0.17)	(2.26)	(5.45)	(2.21)	(11.84)	(2.35)	(1.44)
R^2	0.75	0.28	0.41	0.47	0.63	0.66	0.53	0.37

- Panel B investigates the magnitude of transfer of various welfare programmes in dollar term
- Most of the expenditure is concentrated in permanent retirement and disability benefits ⇒ long-term welfare loss

Calculating Welfare Losses

Two sources of welfare loss:

- Deadweight loss from **governmental transfers**
- Deadweight loss from **involuntary unemployment**

Governmental transfers:

- Calculate increase transfer payment due to China trade shock directly from the regression – $\beta_1 \Delta \widehat{IPW}_{uit}$
- Assume 40% of the increase transfer payment is deadweight loss from the literature (Gruber, 2010)

Involuntary unemployment:

- Neoclassical model of labor-leisure choice
- Deadweight loss arises if foregone wage from unemployment is larger than the value of leisure
- Model calibration comes from the literature (Chetty, 2012)

Total Welfare Losses

In US Dollar per capita:

	1990-2000	**2000-2007**
Loss (Government Transfer)	\$13	\$21
Loss (Unemployment)	\$43	\$69
	\$56	\$90

Total welfare losses from 1990-2007: \$146 per capita

Calculating Welfare Gains (Working Paper Only)

Calculation is based on Arkolakis, Costinot, Rodriguez-Clare, 2012:

- The paper looks at the welfare implication from different trade models
- Aggregate **ex-post** welfare gains can be easily calculated from two parameters:

$$\hat{W}_j = \hat{\lambda}_{jj}^{-1/\epsilon}$$

(1) Changes of the share of domestic expenditure ($\hat{\lambda}$)

(2) Trade Elasticity (ϵ), calibrated from the literature (Simonovska and Waugh, 2011) where $\epsilon \in [-2.5, -10]$

Net Welfare Impacts

	1990-2007
Total Losses	\$146
Total Gains	\$32 ~ \$125
Net Gains from Trade	-\$114 ~ -\$21

- Overall welfare loss from trade with China, very controversial result
- Are the calculations accurate? (e.g. calibration, choice of models, import demand and etc.)
- Other sources of gains? (e.g. dynamic gains from trade)

Main Results

- Chinese import exposure to eight developed countries as IV for Chinese import exposure to the US
- Causal relationship between increasing Chinese import competition and decreasing US manufacturing employment share
- Results are robust against falsification test and additional controls

Labour Market & Public Transfer

- No evidence for changes in working-age population $\Rightarrow$ decline in manufacturing employment is met with higher unemployment and labour force non-participation
- Negative effect on wages across gender and educational level
- Residents without a college degree are more adversely impacted by import competition from China
- Welfare payment is concentrated in retirement and disability benefits $\Rightarrow$ discourages labour participation and long-term welfare loss

Wefare Effects

	1990-2000	2000-2007
Loss (Government Transfer)	\$13	\$21
Loss (Unemployment)	\$43	\$69
	\$56	\$90

	1990-2007
Total Losses	\$146
Total Gains	\$32 $\sim$ \$125
Net Gains from Trade	-\$114 $\sim$ -\$21